# Real-Time Escalator Wrong Direction Detection Using Computer Vision

Final Project Report

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## Abstract

In modern public infrastructures such as metro stations, airports, and shopping malls, escalators play a critical role in transporting large numbers of people efficiently. However, individuals occasionally attempt to walk against the direction of the escalator, which may lead to accidents, congestion, and safety hazards.

This project presents a computer vision based automated monitoring system capable of detecting such anomalous behavior in escalator surveillance videos. The system integrates a lightweight YOLOv8 object detection model to identify persons in video frames, DeepSORT for multi-object tracking, and optical flow analysis to estimate the motion direction of the escalator.

The proposed algorithm compares the trajectory of each tracked individual with the estimated escalator motion vector. If a person moves opposite to the escalator direction, the system classifies the event as **WRONG DIRECTION** and highlights the individual in the output video.

The system processes surveillance footage efficiently and produces annotated video outputs, enabling automated safety monitoring and reducing the need for continuous manual CCTV inspection.

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# 1 Introduction

## 1.1 Problem Statement & Motivation

Escalators are widely used in public transportation hubs, shopping malls, and airports to facilitate vertical movement of people. For safe usage, individuals are expected to move in the same direction as the escalator.

However, in real-world environments, some individuals attempt to walk or run against the escalator direction. This behavior may lead to:

- Accidents and falls
- Congestion in crowded public areas
- Increased safety risks during emergency evacuations

Manual monitoring of CCTV systems is inefficient and prone to human fatigue and errors. Therefore, an automated computer vision based monitoring system is required to detect escalator misuse in real time.

## 1.2 Objectives

The primary objective of this project is to design and implement an automated system that detects anomalous movement on escalators.

Specific objectives include:

- Detect persons in surveillance videos using YOLOv8.
- Track individuals across frames using DeepSORT.
- Estimate escalator motion direction using optical flow analysis.
- Compare human motion vectors with escalator motion vectors.
- Detect and highlight individuals moving in the wrong direction.

## 1.3 Scope of the Project

This project focuses on analyzing escalator surveillance footage and identifying individuals who move against the escalator direction.

The system currently performs:

- Processing of pre-recorded escalator surveillance videos
- Person detection and tracking
- Motion direction estimation
- Visualization of detected anomalies

# 2 Dataset Preparation

## 2.1 Video Data Collection

The dataset used in this project consists of surveillance videos containing escalators in public environments. The videos capture both normal and anomalous behavior.

Examples include:

- People moving along the escalator direction
- People attempting to move against the escalator direction

### 2.2 Data Processing

Before performing anomaly detection, the video frames undergo preprocessing.

The preprocessing steps include:

- Extracting frames from the video stream
- Converting frames to grayscale for motion analysis
- Selecting the Region of Interest (ROI) containing the escalator

## 3 Methodology & Training

### 3.1 Model Architecture

The proposed system integrates three major computer vision components:

- YOLOv8 Object Detection
- DeepSORT Multi-object Tracking
- Optical Flow Motion Analysis

YOLOv8 is used to detect persons in each video frame. The model is designed for high accuracy and real-time performance in object detection tasks.

### 3.2 Tracking using DeepSORT

DeepSORT assigns a unique ID to each detected individual and tracks their position across consecutive frames.

This tracking mechanism allows the system to analyze the trajectory of each person over time.

### 3.3 Optical Flow Motion Estimation

To estimate the motion direction of the escalator, the Farneback Optical Flow algorithm is applied.

Optical flow computes pixel displacement between consecutive frames and generates motion vectors that represent movement direction.

### 3.4 Mathematical Formulation for Anomaly Detection

Let the motion vector of a person be:

$$M_p$$

and the motion vector of the escalator be:

$$M_e$$

The dot product between the vectors is calculated as:

$$D = M_p \cdot M_e$$

If

$$D < 0$$

then the person is moving in the opposite direction of the escalator.

Such cases are classified as anomalies.

## 4 System Architecture

The system pipeline consists of the following stages:

1. Video input acquisition
2. Person detection using YOLOv8
3. Multi-object tracking using DeepSORT
4. Escalator motion estimation using optical flow
5. Motion comparison and anomaly detection
6. Annotated video generation

Figure 1: System Architecture for Escalator Anomaly Detection

## 5 Results & Evaluation

### 5.1 Normal Behavior

When individuals move along the escalator direction, the system labels them as **Normal** and displays green bounding boxes.

### 5.2 Anomalous Behavior

When individuals move opposite to the escalator direction, the system labels them as:
**WRONG DIRECTION**
and highlights them using red bounding boxes.

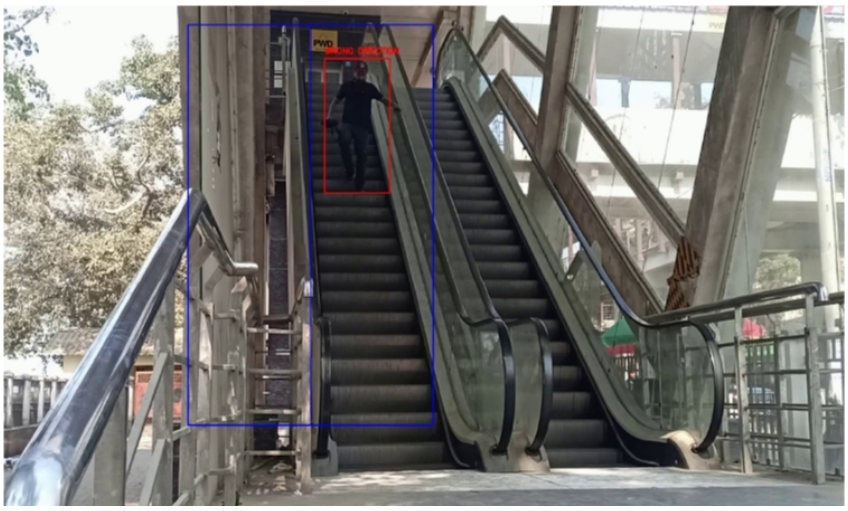


Figure 2: Detected anomaly: person moving against escalator direction

## 6 Conclusion & Future Work

### 6.1 Conclusion

This project demonstrates an automated escalator monitoring system using computer vision techniques. By integrating YOLOv8 detection, DeepSORT tracking, and optical flow motion analysis, the system can effectively detect individuals moving against the escalator direction.

Such systems can significantly improve safety monitoring in crowded public infrastructures.

### 6.2 Future Work

Future improvements may include:

- Real-time deployment with CCTV camera feeds
- Detection of additional anomalous behaviors
- Integration with automatic alert systems
- Deployment on edge AI hardware